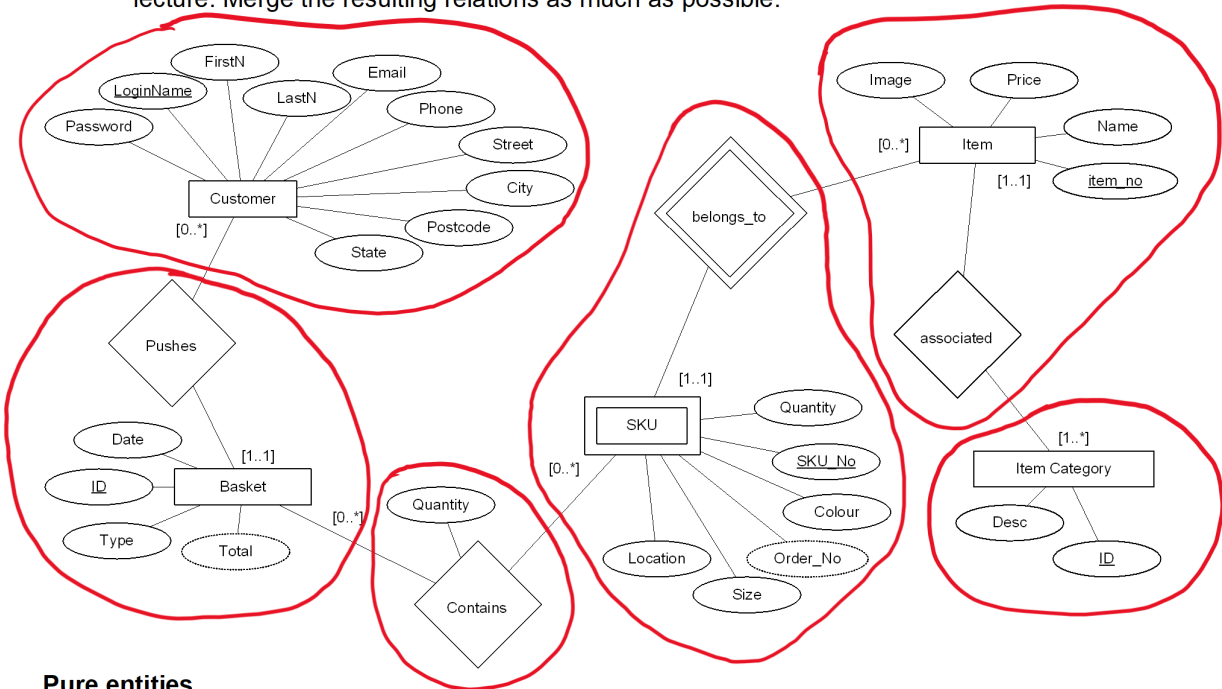


Exercise 09: Transformation into relational schemas

Task 1: Transformation of an ER model into a relational model

- a) Transform the following ER model into a relational model using the rules for mapping from the lecture. Merge the resulting relations as much as possible.



Pure entities

Customer (LoginName, Password, FirstN, LastN, Email, Phone, Street, City, Postcode, State)
Item Category (Desc, ID)

Entities merged with relationships

Basket is merged with Pushes

Basket(ID, Date, Type, LoginName)

LoginName references Customer(LoginName) not null

SKU is merged with belongs_to

SKU(SKU_No, item_no, Quantity, Colour, Size, Location)

item_no references Item(item_no) not null

Item is merged with associated

Item(item_no, Name, Price, Image, Category)

Category references Item Category(ID) not null

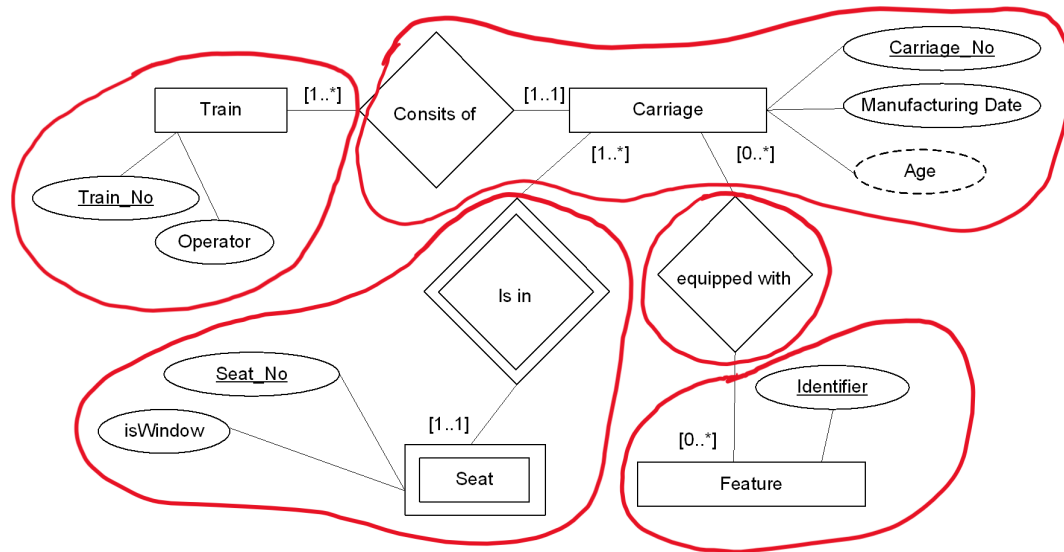
Pure relationships

Contains(Basket, SKU_No, item_no, Quantity)

Basket references Basket(ID)

(SKU_No, item_no) references SKU(SKU_no, item_no)

b) Do the same with the following model.



Pure entities

Train (Train_No, Operator)

Feature (Identifier)

Entities merged with relationships

Carriage is merged with Consists of

Carriage (Carriage_No, Train, Manufacturing Date)

Train references Train(Train_No) not null

Seat is merged with is_in

Seat(Seat_no, Carriage_no, isWindow)

Carriage_No references Carriage(Carriage_No) not null

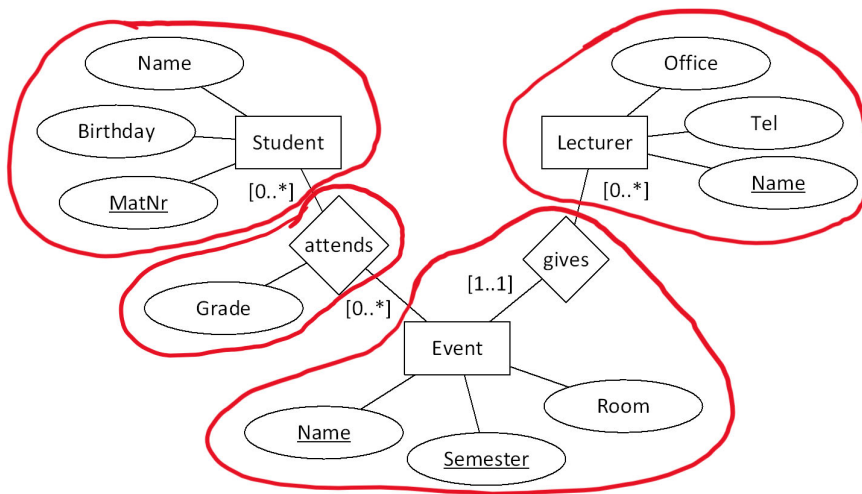
Pure relationships

Equipped with (Feature, Carriage)

Feature references Feature(Identifier)

Carriage references Carriage(Carriage_No)

c) Do the same with the following model.



Pure entities

Student (Name, Birthday, MatNr)

Lecturer (Office, Name, Tel)

Entities merged with relationships

Event is merged with gives

Event(Name, Semester, Room, Lecturer)

Lecturer references Lecturer(Name) not null

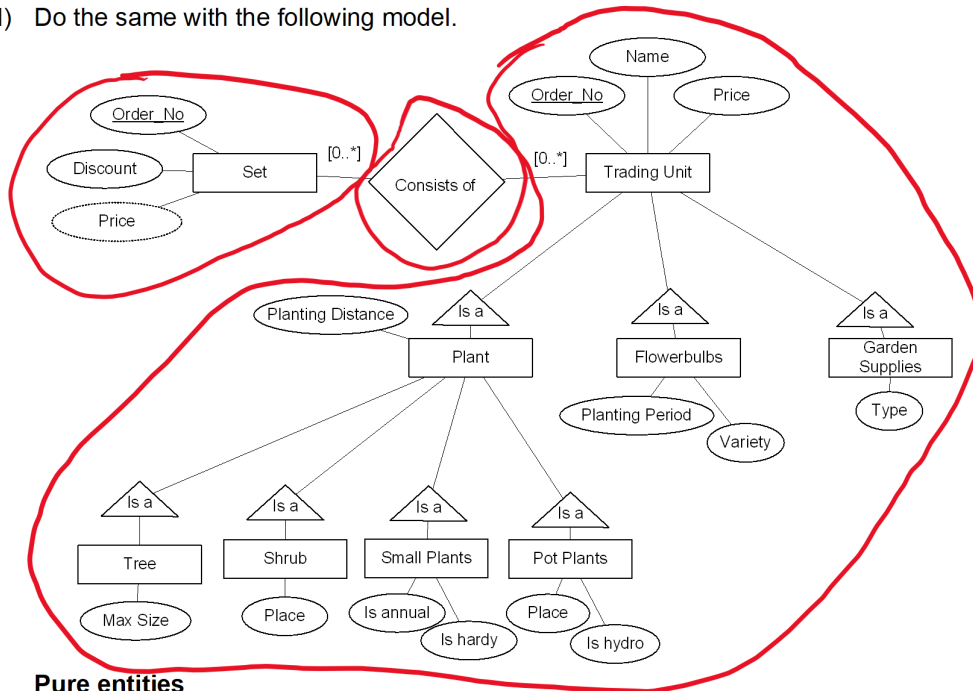
Pure relationships

Attends(Grade, Name, Semester, MatNr)

(Name, Semester) references Event(Name, Semester)

MatNr references Student(MatNr)

d) Do the same with the following model.



Pure entities

Set(Order_No, Discount)

Trading Unit (Order_no, Name, Price, Planting Distance, Planting Period, Variety, Supply_Type, Max Size, Shrub_Type, Is annual, Is hardy, Pot Plant Place, Is hydro))

Entities merged with relationships

None

Pure relationships

Consists of (Set, Trading Unit)

Set references Set(Order_No)

Trading Unit references Trading Unit(Order_No)